



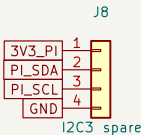
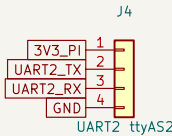
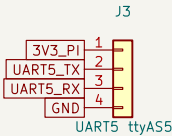
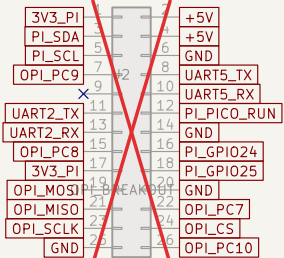
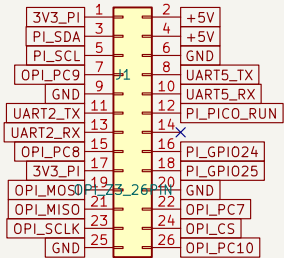
12–48V in; 1500W path stays off-board (external controller + contactor) Pi 4/5 carrier + Pico 2: trolling motor thrust (digipot hijack) + AS5600 servo vanchor-ng Sheet: / File: vanchor-helm.kicad_sch		
Title: Vanchor Helm Board		
Size: A3	Date: 2026–07–01	Rev: A
KiCad E.D.A. 10.0.3	Id: 1/6	



ORANGE PI ZERO 3 CARRIER: J1 male header -> 26-way IDC ribbon to the module

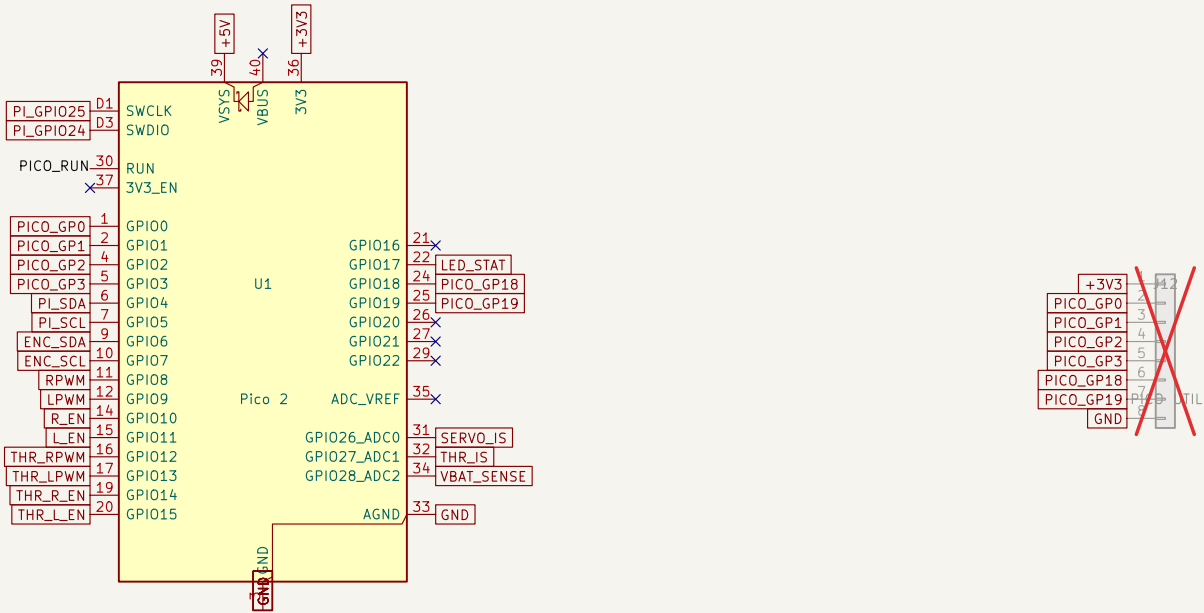
Module is powered through header 5V pins – do NOT also plug USB-C power

UART TX/RX directions per OPI manual – if a device stays silent, swap TX/RX at the JST



PWR_FLAG

MCU: Pico 2 (RP2350) – I2C0 slave @0x42 to Pi, I2C1 master to AS5600
Firmware map: 0x00 CMD {pwm,dir,steer} / 0x10 STATUS {angle,ok,wrap,state,vbat,lsense}; 800ms watchdog
Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).



1k



4.7k



4.7k



47k



10k



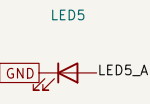
100n



BAT54S



1k



yellow

THRUST: control header for external driver (IBT-2 / BTS7960-class)
J13 pinout matches the common IBT-2 module: RPWM LPWM R_EN L_EN R_IS L_IS VCC GND
Driver held disabled by 100k pull-downs unless the Pico actively enables it
Battery/motor power stays on the external driver – only logic on this board

